

The patient behind the batch

What is ready, and who can decide?

Cell and Gene Therapy (CGT)

A patient-specific journey through collection,
manufacturing and Quality release.



The gaps sit between systems

153 / 499

legacy-ready records
lacked a QMS release

Synthetic data contradiction.
Not evidence of a real-world infusion.

Where the handoff breaks	What we investigate
Conflicting identity	Source, owner and authority
Missing release evidence	Policy and readiness rules
Uncertain reservation	Timeout, retry and reconciliation

Output: current-state process, C4 views, root causes and a baseline.

A clear job for the control tower

The coordinator’s job

Know what is ready.
See what blocks it.
Reach the right owner.

Shared domain language
Patient, Batch, Evidence,
Decision and Exception

Product need	Required behavior
Trustworthy status	Show satisfied, not satisfied or unknown
Traceable facts	Keep source and time with the evidence
Accountable action	Name the exception and decision owner
Safe scope	Qualify data use and human authority

AI must earn a place

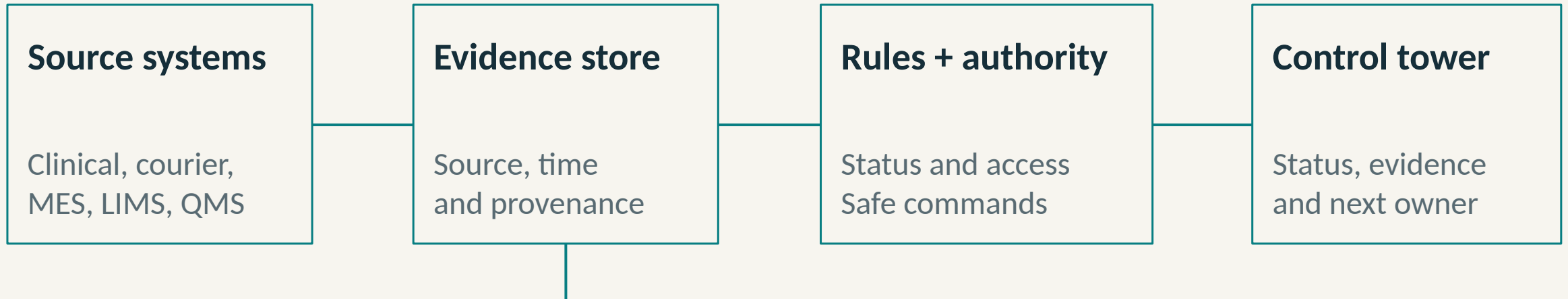
Option	Suitable job	Capstone position
Conventional automation	Rules, authority and safe commands	Selected core
Bounded AI assistant	Explain a permitted evidence packet	Optional, value unproven
Autonomous agent	Choose and execute consequential actions	Not selected

AI off works today

Future AI must improve task quality, latency and total cost within agreed limits.

Evidence becomes a decision

Operational evidence path



Optional assistant

Reads a permitted evidence packet.
Offers an explanation. Cannot authorize actions.

Executive architecture view

The contract makes control testable

Control boundary	Engineering contract	Evidence to retain
Identity and release	Authorized human decision	Role checks and audit record
Uncertain commands	Reconcile before retry	Idempotency and fault tests
Optional AI	Bounded context and tools	Grounding and fallback tests
Enterprise operation	Security and supplier controls	ADRs, threat model and exit plan

The PRD and delivery specification link requirements to code, tests and proof.

Three moments prove the design

01

Conflict

Two sources disagree
about patient identity.

**Human resolves identity.
Readiness recalculates.**

02

Silence

The reservation succeeds.
Its reply times out.

**Reconcile the outcome.
Prevent another dispatch.**

03

Hold

The evidence is assembled.
Release still waits.

**Quality authorizes release.
Retain the evidence.**

Artifacts inform the PRD. Code implements it. The app demonstrates it.

A demo earns the next test

29 / 31

requirements internally verified

Production-verified requirements: 0

Evaluation record	Scope
57 cases executed	55 pass, 2 inconclusive
16 original cases	7 full, 9 partial assertions
39 extension cases	Structural probes only

Next: studies, load, live integrations,
independent review and operating readiness.

Current use: synthetic academic demo. External readiness gates remain open.

Value must survive daily use

Business measure	Supplied baseline	Observed production after
Journeys needing reconciliation	31.5%	Not yet measured
Manual status updates / journey	12.7	Not yet measured
Complete cross-system traceability	82.4%	Not yet measured

Daily operating proof

Availability, recovery, drift,
overrides and cost per successful outcome.

Supplied synthetic baselines.
Definitions need validation before
any before/after benefit claim.

Our decision: advance in controlled steps

Accept the academic capstone. Fund the next evidence gates.

DAY 30

Confirm owners,
policy and baseline

Signed mandate

Current decision: restrict and change

DAY 60

Test source contracts
and controlled rules

Reviewed sandbox evidence

DAY 90

Studies and recovery
Independent review

A decision on pilot scope

If gates fail: restrict, suspend or retire.
Revoke access and retain the lessons.